

Retrieval Augmented Generation (RAG)

Project Report

Field: Artificial Intelligence / NLP

Abstract

Retrieval Augmented Generation (RAG) are powerful artificial intelligence systems trained on vast amounts of text data. They are designed to understand, generate, and analyze natural language. These models use deep learning architectures, particularly transformer networks, to capture contextual relationships between words. RAG s are widely applied in Natural Language Processing tasks such as text summarization, translation, question answering, sentiment analysis, and conversational AI systems. Organizations integrate RAG s into automation platforms, intelligent chatbots, recommendation systems, and digital assistants to improve efficiency and user experience. Despite their advantages, RAG present challenges including computational cost, bias in training data, and ethical considerations in AI deployment.

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Introduction

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Methodology (Project-Based)

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Implementation / Discussion

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Advantages & Disadvantages

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Applications

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Future Scope

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NLP Pipeline

Input Data → Transformer Model → Generated Output

Used for NLP Tasks, Chatbots, and Automation

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Introduction

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Literature Review

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Methodology (Project-Based)

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Implementation / Discussion

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Advantages & Disadvantages

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Applications

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Future Scope

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Chatbot Workflow

Input Data → Transformer Model → Generated Output

Used for NLP Tasks, Chatbots, and Automation

Conclusion

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Introduction

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Methodology (Project-Based)

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Implementation / Discussion

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Advantages & Disadvantages

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Applications

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Automation System

Input Data → Transformer Model → Generated Output

Used for NLP Tasks, Chatbots, and Automation

Future Scope

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Literature Review

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Methodology (Project-Based)

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Implementation / Discussion

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Advantages & Disadvantages

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LLM Architecture

Input Data → Transformer Model → Generated Output

Used for NLP Tasks, Chatbots, and Automation

Applications

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Future Scope

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